

Removable Complete and Partial Dentures

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5/19/2026

Complete denture



A complete denture replaces all missing teeth in the upper or lower jaw

Partial denture



A removable dental prosthesis used to replace one or more missing teeth in an arch where some natural teeth still remain.

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It restores masticatory function, aesthetics, and prevents the remaining teeth from shifting.

Acrylic base

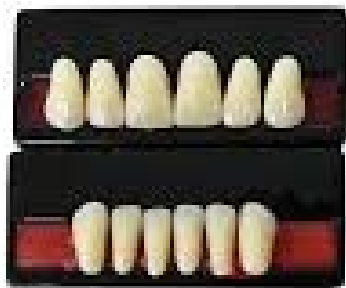


the pink-colored resin part of a denture that mimics natural gingival tissue.

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It serves as the foundation that holds the artificial teeth in place and distributes occlusal forces during chewing. In partial dentures, it can be used alone or combined with a metal framework for added strength

Artificial Teeth

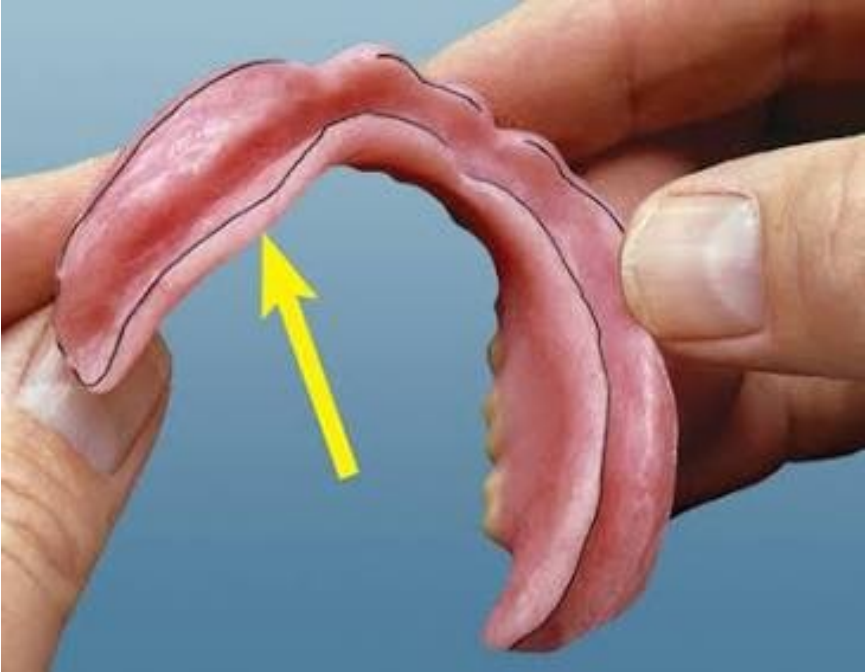


manufactured prosthetic teeth made of acrylic resin or porcelain. Selected by shape, size, and shade.

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acrylic versions are highly preferred in dental technology due to their strong chemical bonding with the denture base and lower risk of fracturing.

Denture Flange



the vertical extension of the denture
base

that runs from the necks of the
artificial teeth down into the
sulcus.

Create a proper peripheral seal for
optimal denture retention

Palatal Plate



the major connector of an upper partial or complete denture that covers all or part of the hard palate.

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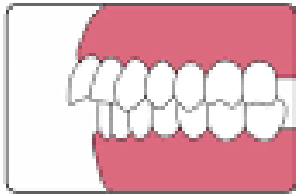
it can be made of acrylic resin or a thin, cast metal alloy (like cobalt-chromium).



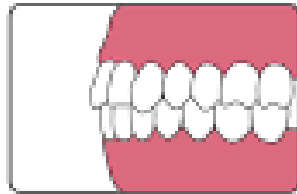
Occlusion



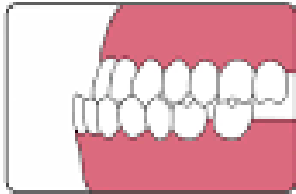
Types of Occlusion



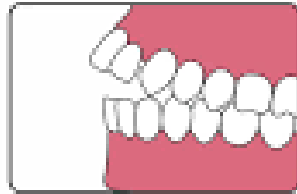
Healthy



Overbite



Underbite



Openbite

Refers to the static and dynamic contact relationship between the maxillary (upper) and mandibular (lower) teeth when they meet during rest or functional movements.

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In dental technology, achieving proper occlusion is critical for balanced chewing forces, protecting the temporomandibular joint (TMJ)

Bite registration



the process of recording the exact relational position of the upper and lower teeth or jaws in a stable medium (such as wax, or bite silicone).

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This accurate 3D record is transferred to a dental articulator to duplicate the jaw movements in the lab, ensuring the final prosthesis has correct occlusion and vertical dimension.

Metal Framework



The metal framework provides strength and support in partial dentures.



Clasp

A clasp helps hold the partial denture in place.

holding the removable appliance securely in place and preventing it from lifting during eating or speaking.

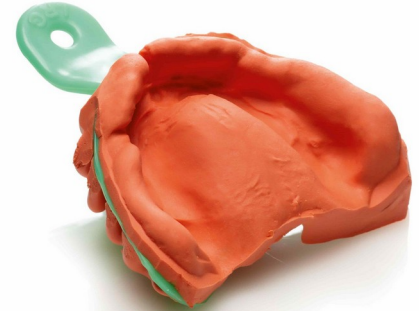


Preliminary Impression



also called an initial or diagnostic impression, is a negative likeness of the patient's oral tissues, usually taken using alginate in a stock tray.

It is the first impression step, used in the laboratory to pour diagnostic plaster models, plan treatment, and fabricate custom impression trays for the final, precise impression.



Wax Rim



help determine the correct bite and tooth position. A raised wax border built onto a temporary denture base to occupy the space of the missing teeth. It is used by the dentist during the bite registration phase to determine the patient's vertical dimension, center line, smile line, and the correct positioning for the future artificial teeth.

Flasking

Steps of processing of complete denture



The laboratory process of investing (encasing) a dental prosthesis in a sectional metal or plastic box (called a flask) using dental plaster or stone.

This technique creates a precise, heat-resistant mold around the wax-up model, allowing the temporary wax to be boiled out and replaced with permanent acrylic resin.



Finishing and Polishing

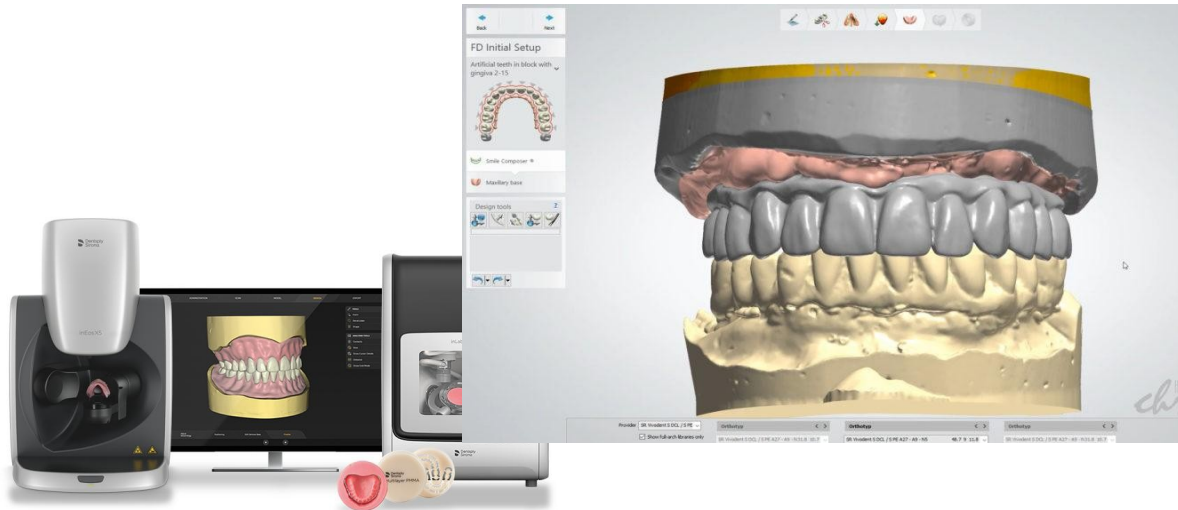


improve the comfort and appearance of the denture. the final laboratory phase where the processed acrylic or metal denture is refined, smoothed, and shined.

Finishing involves using burs, stones, and scrapers to remove excess material and shape the anatomy, while polishing uses pumice, rag wheels, and high-shine compounds to create a mirror-like surface.

CAD/CAM Denture

CAD/CAM technology allows digital design and manufacturing of dentures.



Polymerization



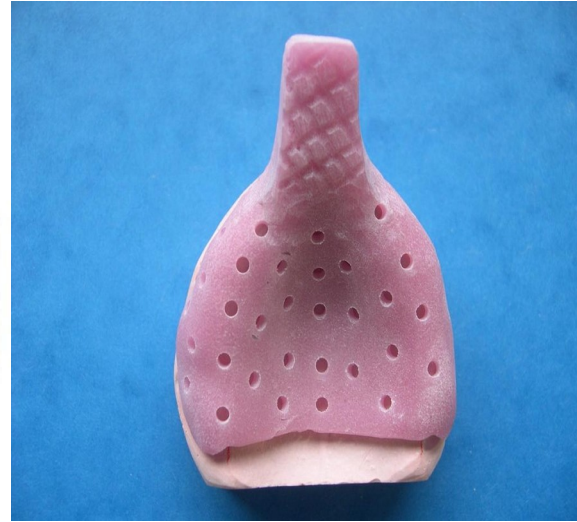
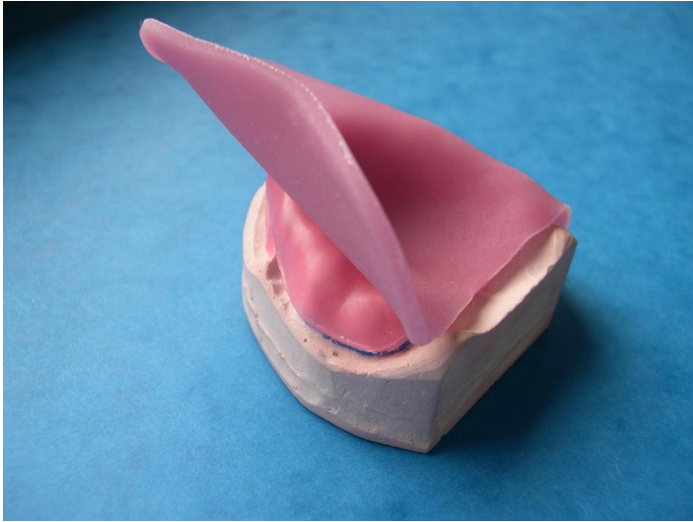
Chemical reaction where small molecules (monomers) convert into a large chain network (polymer), turning the soft acrylic dough into a hard material.

During polymerization, heat-cured acrylic resin is processed inside a flask to produce a strong denture base.



Custom Tray

specially made impression tray used for final impressions in denture fabrication.



Articulator

mechanical device used to simulate jaw movements during denture fabrication



3D Printed Denture

digitally designed and manufactured denture, using additive 3D printing.

